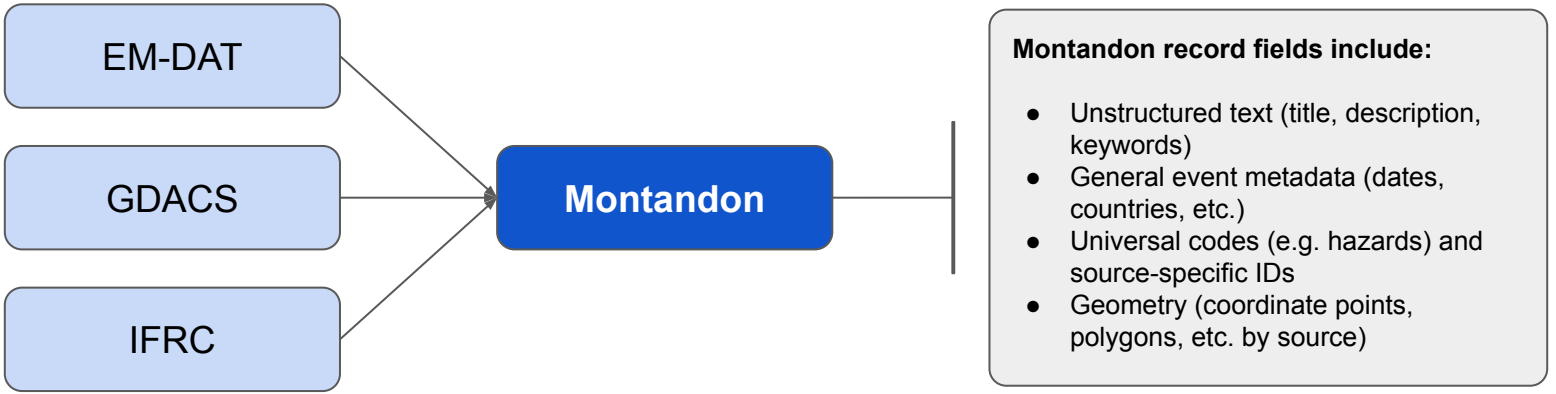
Exploring applications for the Montandon Global Crisis Data Bank

Jeongmin An, Jehan Bugli, and Aidan Carlisle

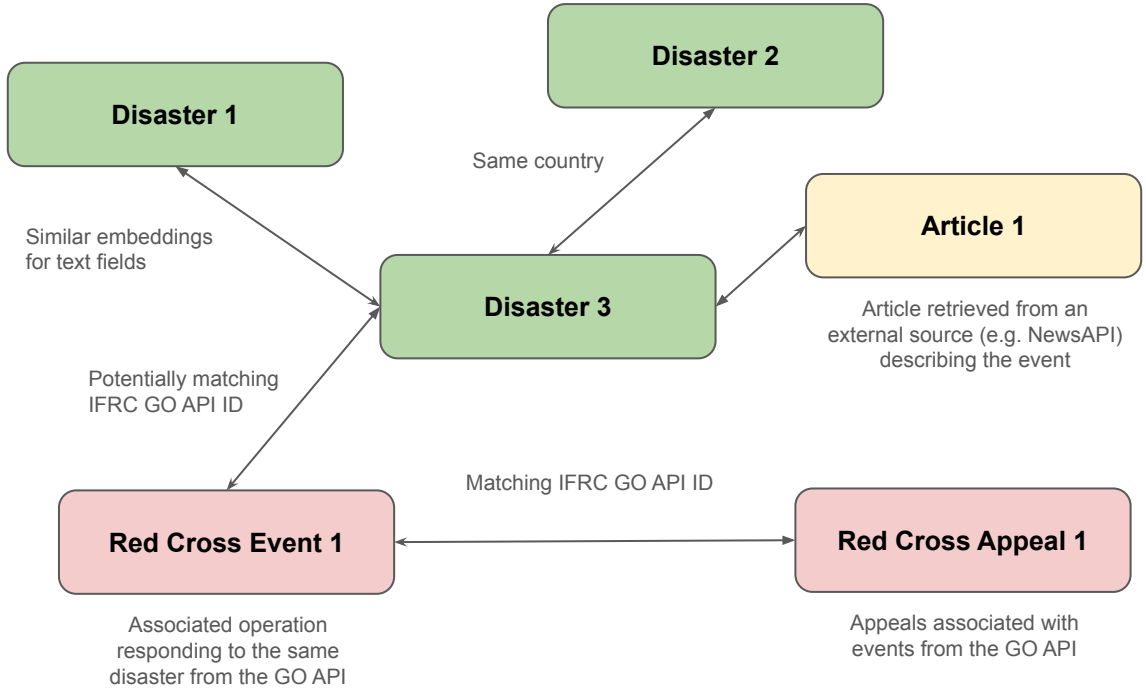
The Montandon Data Bank collects disaster data from disparate sources, exposing it under a unified system

- **Key research question:** how can the data bank be used to understand disasters and respond to them more effectively?



We built a network connecting Montandon records and external data to serve as an LLM query layer

- Montandon records are connected to each other, external articles, and the IFRC GO API
- Temperature data is collected from the NASA POWER API and integrated with relevant Montandon records
- Nodes and edges are saved in a local Neo4j instance
- LLMs receive tools that execute templated Cypher queries



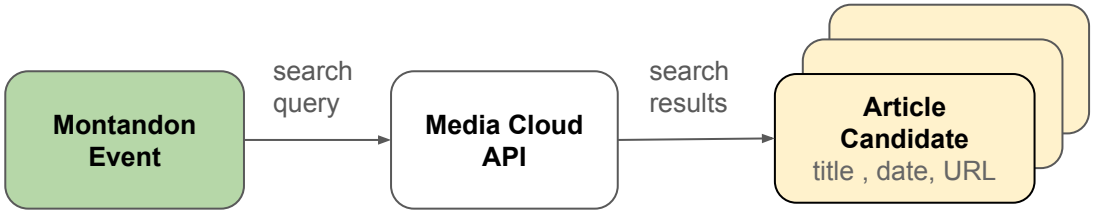
Historical disaster coverage requires a searchable news archive

- NewsAPI.org Developer access searches only recent articles.
- **Media Cloud** provides historical article discovery for our event dates.
- **We search by event date, hazard, and country;** matches are candidates, not confirmed evidence.

Pilot: 2025 Texas flash flood

July 3-10, 2025

“Flash Flooding” AND “United States” AND language:en



Observed pilot: 100 article candidates returned in the initial API request; relevance not yet verified.

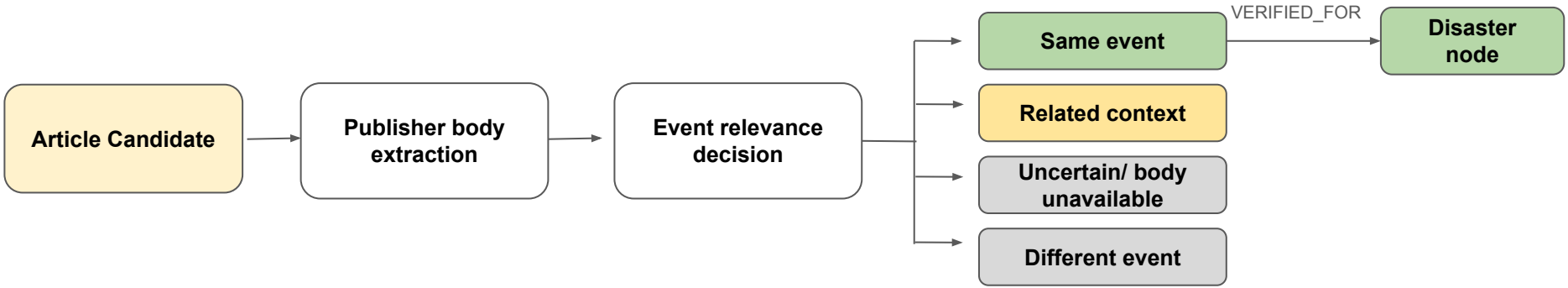
Article candidates must be checked before linking them to disaster

Search results mix signal and noise

Looks relevant	Texas flash flood is a preview of chaos to come
Looks unrelated	EPA article about contrails

Both appeared in the first 100 results for the same event query.

- Media Cloud supplies title, date, and URL - not downloadable article text.
- We fetch publisher HTML when accessible and extract the body
- A four-way decision compares article evidence with the event.



Local LLM performance was benchmarked against a set of fact retrieval questions

- We created question banks with known answers from the network
- LLMs were given these questions and graded on performance (% correct)
- Answers were requested in specific formats supporting substring checks
- Testing centered around Qwen3-1.7B, Qwen3-5-4B, and Qwen3.5-9B

“Which disaster (by starting date) had the highest wind speed?



<answer>2025-10-01</answer>



Disasters on 2025-10-01 and 2025-10-09 had equivalent wind speed.

“How many people were affected by X disaster”?



X disaster affected **12,050** people.

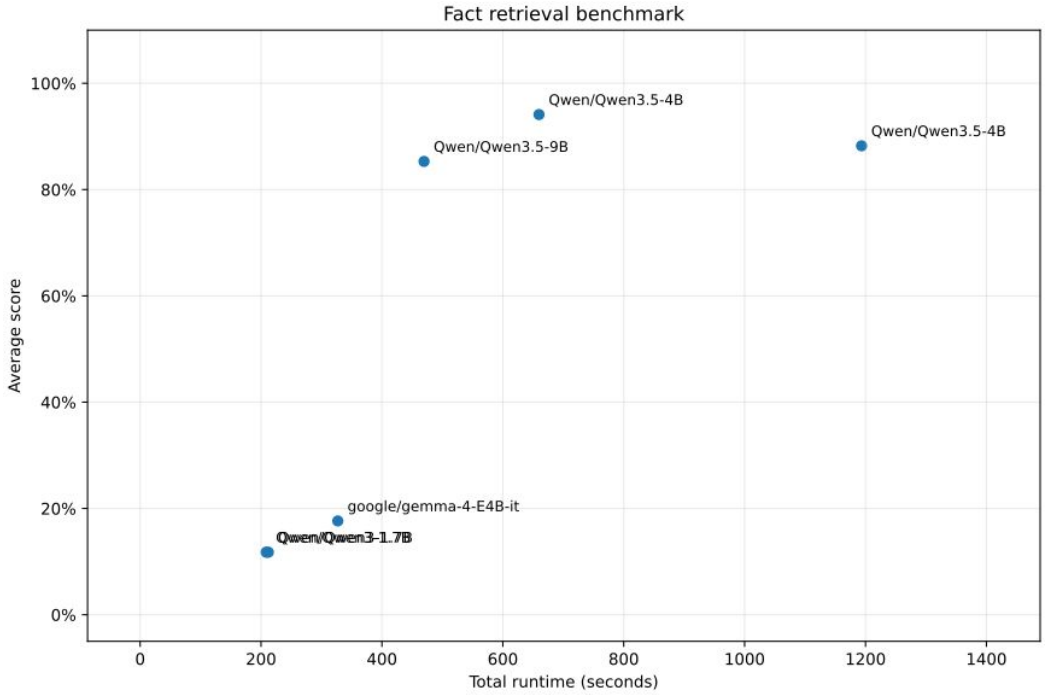


X disaster affected 20 people.

Qwen3.5-4B performed the best in evaluations, making that our local LLM of choice for an initial demo (1/2)

- Interestingly, Qwen3.5-4B performed better than its 9B counterpart
- All models had some difficulty following instructions well
- Benchmarks informed the templated tool approach due to difficulties with raw Cypher
- General fact retrieval was tested alongside historical weather data (latter with/without tools) to investigate innate world knowledge

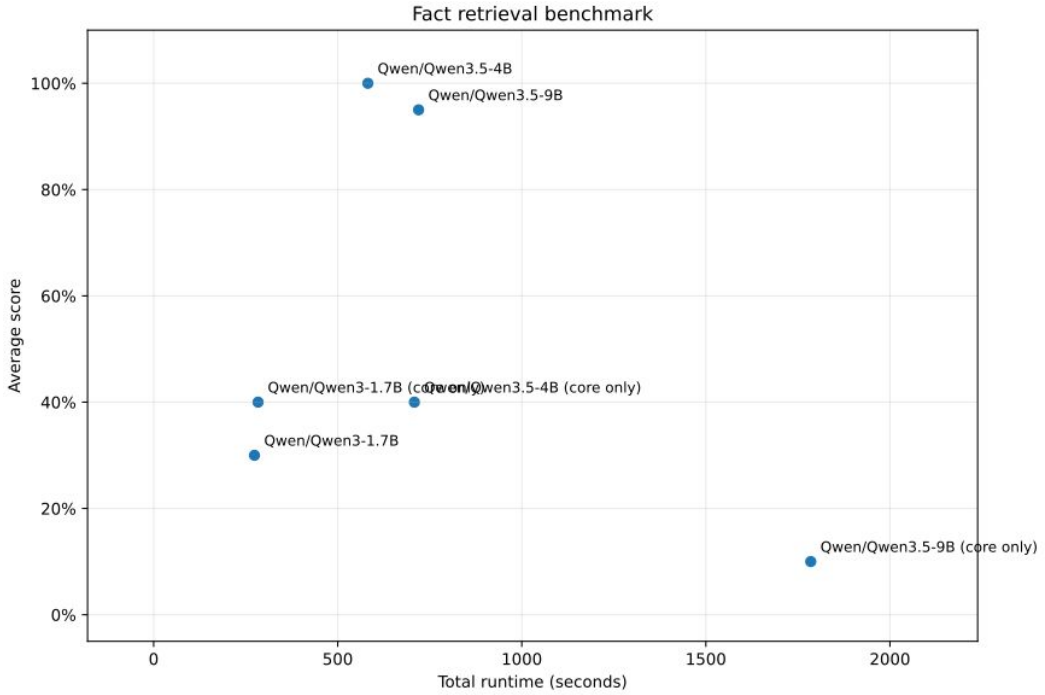
Benchmark 1: General fact retrieval



Qwen3.5-4B performed the best in evaluations, making that our local LLM of choice for an initial demo (2/2)

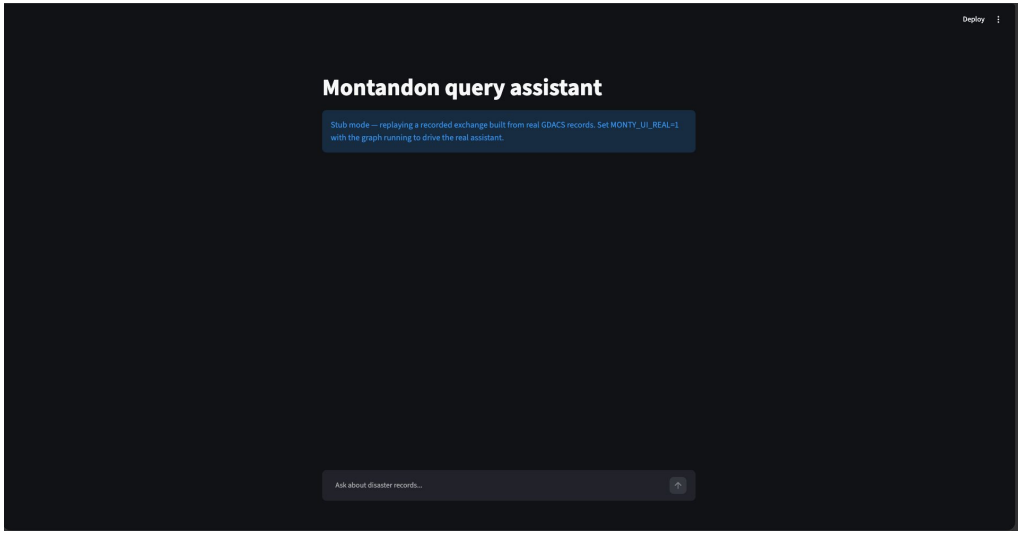
- Interestingly, Qwen3.5-4B performed better than its 9B counterpart
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- Benchmarks informed the templated tool approach due to difficulties with raw Cypher
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Benchmark 2: Historical weather data



User Interface

- Built a Streamlit app that wraps the QueryAssistant and shows its output
- Shows conversation history and a list of specific disaster records that contributed to the answer



Deploy

Give me floods and impacts in Spain

search_disaster_events — 1 row(s)

arguments

```
{
  "country_code": "ESP",
  "hazard_code": "MH0600"
}
```

returned

event_id	title	start_datetime	end_datetime	country_codes	hazard_cc
gdacs-event-1104151-1	Flood in Spain	2026-09-09T01:00:00Z	2026-09-11T01:00:00Z	ESP	FL, MH060 nat-hyd-fl

get_disaster_context — 1 row(s)

arguments

```
{
  "event_id": "gdacs-event-1104151-1"
}
```

returned

Ask about disaster records...